For each of the following systems, find out how many solutions they have. Check also if there are redundant equations that can be removed.

Hint In order to identify the rank of the matrix, you may want to transform it into a row echelon form using elementary row operations. A matrix is in row echelon form when it satisfies the following conditions:

- each leading entry is in a column to the right of the leading entry in the previous row,
- rows with all zero elements, if any, are below rows having a non-zero element.

Some definitions also impose that the first non-zero element in each row is 1. However, this conditions is not needed to obtain the rank of the matrix.

Question 1

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 0, \\ 4x_1 + 5x_2 + 6x_3 &= 0, \\ 7x_1 + 8x_2 &= 0. \end{aligned}$$

Question 2

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 0, \\ 4x_1 + 5x_2 + 6x_3 &= 0, \\ 7x_1 + 8x_2 + 9x_3 &= 0. \end{aligned}$$

Question 3

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 0, \\ 2x_1 + 4x_2 + 6x_3 &= 0, \\ 5x_1 + 10x_2 + 15x_3 &= 0. \end{aligned}$$

Question 4

$$\begin{aligned} x_1 + x_2 + x_3 &= 1, \\ x_1 - x_2 + x_4 &= 1, \\ x_1 - 5x_2 - 2x_3 + 3x_4 &= 1. \end{aligned}$$

Question 5

$$\begin{aligned} x_1 + x_2 &= 0, \\ x_1 + 2x_2 + 2x_3 &= 0, \\ 3x_2 + 3x_3 &= 0. \end{aligned}$$